

1.小车红外巡线

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1.学习目标

通过四路巡线模块与小车结合，实现小车巡黑线。

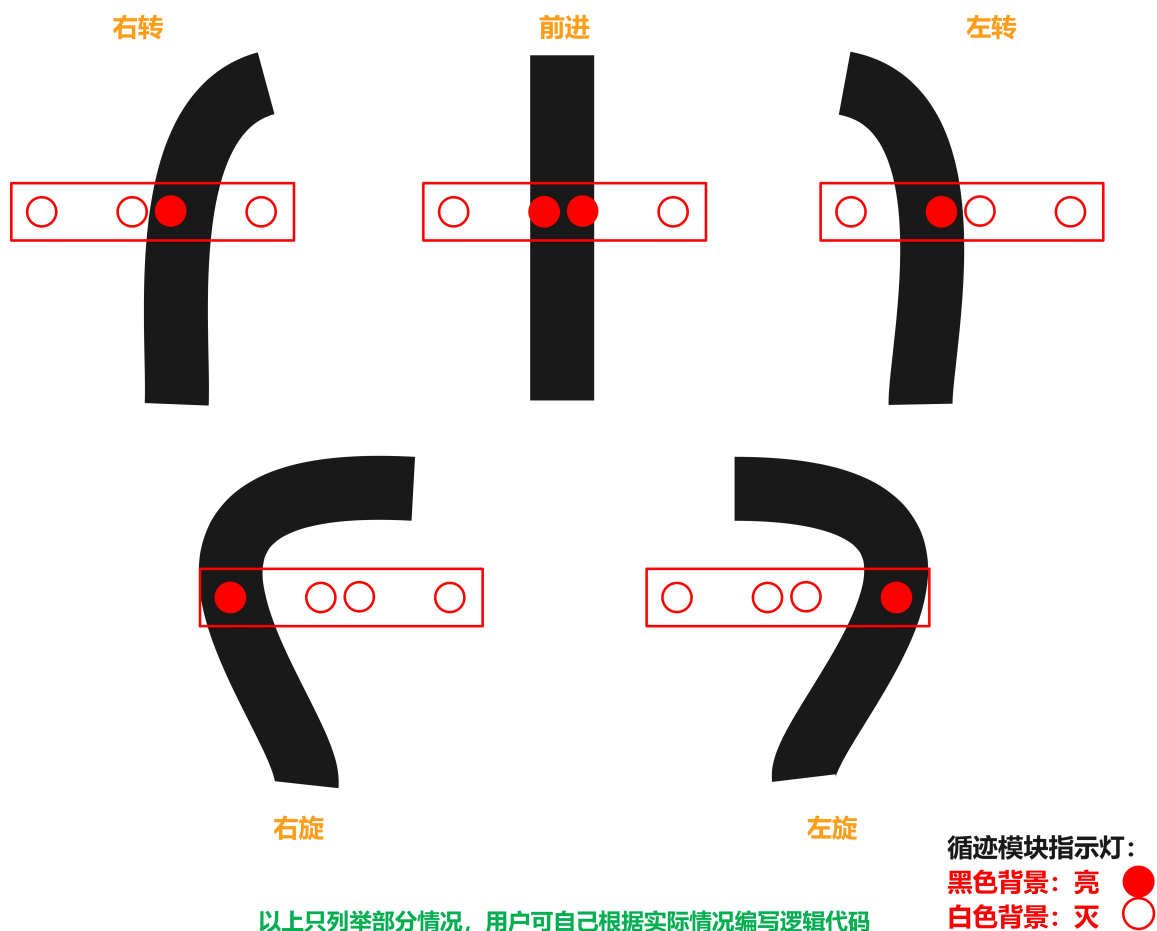
2.实验准备

1.小车接线已安装

2.调试四路巡迹模块。调节四路循迹模块旋钮，调节成遇黑线时，指示灯亮；不是黑线时，指示灯灭。

3.实现原理

检测不同形状的赛道时，四路巡线传感器的状态，如下图所示。根据传感器状态来控制小车进行不同的运动。



4.代码分析

源代码路径:

/home/pi/project_demo/05.Comprehensive_gameplay/1.infrared_patrol_line.ipynb

```
import sys
sys.path.append('/home/pi/project_demo/lib')
#导入麦克纳姆小车驱动库 Import Mecanum Car Driver Library
from McLumk_wheel_Sports import *

speed =30#30
try:
    while True:
        # 从I2C读取巡线传感器数据 Read line sensor data from I2C
        track_data = bot.read_data_array(0x0a, 1)
        track = int(track_data[0])

        # 解析巡线传感器的状态 Analyze the status of the line patrol sensor
        x1 = (track >> 3) & 0x01
        x2 = (track >> 2) & 0x01
        x3 = (track >> 1) & 0x01
        x4 = track & 0x01
        """
        x2 x1 x3 x4
        | | | |
        L1 L2 R1 R2
        """
        lineL1=x2
        lineL2=x1
        lineR1=x3
        lineR2=x4

        if lineL1 == 0 and lineL2 == 0 and lineR1 == 0 and lineR2 == 0: # 都是黑色, 加速前进 All black, speed up
            print("1")
            print(lineL1,lineL2,lineR1,lineR2)
            move_forward(int(speed))
        elif( (lineL2 == 0 or lineL1 == 0) and lineR2 == 0):#右锐角: 右大弯,0表示检测到黑线 Right acute angle: right big bend, 0 means black line is detected
            print("2")
            print(lineL1,lineL2,lineR1,lineR2)
            rotate_right(speed)
            time.sleep(0.05)
        elif lineL1 == 0 and (lineR2 == 0 or lineR1 == 0): # 左锐角或左大弯 Left sharp angle or left sharp bend
            print("3")
            print(lineL1,lineL2,lineR1,lineR2)
            rotate_left(int(speed*1.5)) # 左急转弯 Sharp left turn
            time.sleep(0.15)
        elif lineL1 == 0: # 左最外侧检测 Left outermost detection
            print("4")
            print(lineL1,lineL2,lineR1,lineR2)
            rotate_left(speed) # 左急转弯 Sharp left turn
            time.sleep(0.02)
        elif lineR2 == 0: # 右最外侧检测 Right outermost detection
            print("5")
            print(lineL1,lineL2,lineR1,lineR2)
```

```

        rotate_right(speed)
        time.sleep(0.01)
    elif lineL2 == 0 and lineR1 == 1: # 中间黑线上的传感器微调车左转 The sensor
on the middle black line fine-tunes the car to turn left
        print("6")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_left(int(speed)) # 左转 Turn left
    elif lineL2 == 1 and lineR1 == 0: # 中间黑线上的传感器微调车右转 The sensor
on the middle black line fine-tunes the car to turn right
        print("7")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_right(int(speed)) #右转 Turn right
    elif lineL2 == 0 and lineR1 == 0: # 都是黑色，加速前进 All black, speed up
        print("8")
        print(lineL1,lineL2,lineR1,lineR2)
        move_forward(speed)

    # 等待一段时间再进行下一次检测 wait for a while before the next test
    time.sleep(0.01)

except KeyboardInterrupt:
    # 当用户中断程序时，确保所有电机停止 Ensure that all motors stop when the user
interrupts the program
    stop()
    print("Ending")

```

5.实验效果

我们把小车放在巡线赛道上，确认已经调节了模块旋钮，调成遇黑线时，指示灯亮后，运行代码块，小车会巡黑线行驶。